1	Using environmental sensors (air pressure, temperature, luminance, volatile organic compounds, etc.) to control parameters within algorithms that drive synthesizers as a way of creating environmentally-responsive installations and software patches.
6	I have built new drums!
7	I have applied digital signal processing on an ESP32 microcontroller to modify and electric guitar. I have done the signal processing with my own implementation of my code. It also has a configurable touch panel that you can experiment with or seamlessly switch from filters that you've configured in advance
8	I built a sample instrument plugin with JUCE that supports native Ambisonic sample playback.
9	iOS/C++, Python, and Matlab scripts to cut up/modify samples (both real-time and offline processing)
10	I build games for practicing and performing simple music
11	Used logic pro synthesizers and plug-ins to create sounds Fine-tuned transformer model on a small dataset to generate music
13	audio plugin instruments, max/msp patches
14	i made a music instrument from stretchy fabric, built some oscillator based noisemakers
15	I regularly build Supercollider patches/instruments to serve my artistic necessities. I have also used other software and general-purpose programming languages for the realisation of additional more generic tools (e.g. a library for real-time audio synthesis in Unity), as well as generative artworks distributed as software (e.g., .exe, .apk)
16	Autotune. Guitar tone clone
20	The tool didn’t incorporate legacy file formats so I had to build a converter
21	I have created software that can generate music improvisations or analyse music, which I have done for research, but not for my music practice
22	I have modified my HSS Squire Affinity Stratocaster in many ways: I stripped off the finish and replaced it with a polyurethan wood stain, added a
23	Add control mechanisms to pre-trained music transformers
24	I've built a sculptural soundscape improvisation instrument, using a number of telescoping antennae and distance sensors to control processing parameters in a max/MSP patch. This included circuit and enclosure fabrication, 3D printing, Arduino coding, and max/MSP patching.
25	I have made Arduino-based music instruments that include pressure sensors, inertial measurement units, light sensors, proximity sensors, etc. These were integrated with Max on a laptop. I do not, however, regularly use these in my music practice, which is separate.
27	I designed a live coding environment called Facet: https://github.com/mjcella/facet I designed a web application for flow-based audio programming called Wax: https://wax.bz/ I have built approximately 10 open-ended Max for Live devices for generative music: https://nnirror.gumroad.com/ I've also built several prototype-level hardware instruments, listed below. 1. Antennaesthesia Antennaesthesia is a modified helmet with an antenna probe gives its wearer and others in the room the ability to see, hear, and feel the hidden electromagnetic fields that surround us in everyday objects. Materials used: Electro-Smith Daisy Seed, bicycle helmet, gooseneck microphone stand, inductors, Arduino, ws2812 programmable LED strip, contact microphones, speakers, op-amps. 2. Octile Octile is a polyphonic, rotating cylindrical cousin of the ribbon controller. There are 8 ribbon controllers made of reel-to-reel tape, wrapped around the cylindrical surface which can rotate bi-directionally via a foot controller. Inside the cylinder is a circuit powered by a 9V battery, connected to 8 oscillator chips. Each oscillator is running at a unique frequency and is connected to one of the 8 tape strips. By using gloves with conductive touch points connected via patch cable to Max, and measuring the frequency and amplitude of all 8 input signals, it is possible to determine which tape strips you're touching and where. Materials used: 555 oscillator chips, reel-to-reel tape, 9V battery, Max, plastic bucket, 12V DC motor, DIY turntable, modular synthesizer, gloves, conductive tape. 3. Breathing Then and Now Breathing Then and Now is a mixed media composition which manifests the immensity of geological time by rescaling it into a human-perceptible frame of reference through sound. Materials used: Petoskey stones, a small aquarium, contact microphones, water, two PuckJS microcontrollers embedded into Petoskey stones, Max, NodeJS, WebSockets, and a custom-trained neural network based on BBC sound effects.
30	I am interested in using technologies in a way that breaks them. The same as true for “analog instruments“. (think popping balloons or deliberately, putting too much data into an instrument.)

31	I created an instrument using Kontakt 5 by recording and mapping drill sounds. I also made an attempt to recreate the Stam Audio SA-2A with deep learning, but I wasn't able to get satisfactory results.
34	Kontakt scripts editing, created M4L Devices, physical alterations to winds and percussion instruments
37	I have made a handful of custom instruments/tools over the years. The most recent example was a physical mixer for spatial audio. Using some magnets sandwiched around an acrylic dome, an under-mounted camera, and the spat library for Max MSP, I was able to create a tool where a user moves a magnet around the dome to change it's position in a spatial audio mix. This worked pretty well for orchestrating specific spatial trajectories in real time.
39	mostly built max patches
40	I have already built guitars from scratch, re-did the wiring on many others. I wrote some audio software and customized my daws and controllers
43	ProTools has many ways to create and alter sounds.
44	I am currently working towards using python to classify various gestures and then creating custom Max Patches to integrate the outputs with regular sound patches. Also, I have made custom Javascript patch to get accelerometer data streamed from phone using TCP/IP whereas by default MaxMSP supports UDP
46	i use software filters to personalize timbres
51	Made many modular synth patches, don't know if that counts.
52	Made a theremin, modified a guitar effect pedal, and programmed a VST in C++ using the Juce framework (it worked, but was quite difficult and frustrating)
53	See www.alternativemeasures.app
54	Built filters for analog synth 1971, built electronic/ computing instruments since 98
55	I don't just hack or build tools—I design and engineer recursive, modular systems that redefine the nature of authorship and performance. This began with the co-development of the Virtual Orchestra (US Patent 6696631), a real-time digital performance system that replaced or enhanced traditional pit orchestras and sparked global controversy. It challenged deep assumptions about authenticity, labor, and the role of machine collaboration in live performance. The system has been deployed by some of the world's largest producing organizations—including Cirque du Soleil, Cameron Mackintosh Ltd., and numerous Broadway and opera houses—bringing machine agency into environments previously dominated by human ensembles. That project reframed not only how we create music, but who or what is allowed to participate in its creation. Since then, I've developed many systems, including AI-driven media control (via API + OSC + QLab), quantum computational systems that attempt to solve for Mozart (employing Qiskit and IBM Quantum), and embedded meta-gamelan ensembles within interactive shadow puppet theatre. These systems exist within canonized fictive environments—recursive, rule-bound worlds where aesthetic logic and computational agency intersect to create a larger structural consistency across invented genres that honor world tradition but extend into different environments in the same manner as a LoR or GoT or Star Wars create a convincing and consistent self-contained. universe. Because of this, I must question the very term
61	I have built many instruments and plugins within my PAT 204 class such as a synthesizer, pedal board, vocoder, and a pitch tracker.
62	Simple waveform generators in softwares such as Max; DIY Theremin as part of a class project
66	For example, I personalized the Lyle Mays' ocarina sound with syx plugin.
72	Design / build eurorack modules, create VR music groovebox, generative pure data patches. All focusing on improvisation and performance control
73	I just build software toolkits that do what I want -- usually pseudorandomized delays and filters, that kind of thing
74	Often I use found objects to create DIY percussion instruments.
76	Modifying SC and/or Max patches; reworking turntables for contact mics and wire; built modular synth modules; etc.
77	building synth patches and rudimentary max patches
80	I use a no-input mixer, a customized MIDI controller, and a customized electromagnetic pickup.

82	I hacked a noise maker to add a speed adjustment knob. I turned a you piano into a midi keyboard.
83	Marimba, Nord Stage, Kontakt, Ableton instruments
84	Norns is supercollider driven with Lua, I've build scripts for that. Similarly "crow" is a lua scriptable eurorack module and I've created a modular instrument driven with it. Additionally lots of soldering/DIY/circuitbending.
85	I didn't build it personally (coding) but I am currently developing a flexible visual media playback tool with my sister and collaborator that uses a score follower to allow pre-choreographed visual media to flex and follow live performer's rubato without click tracks or manual cues.
88	Using webcams as instruments
89	Most recently, I have built piezoelectric devices to trigger samples from a sampler when they physically engaged with
90	I have made instruments out of trash and helped kids do the same at events
91	I regularly take things apart and hack them to get cool sounds. Have built instruments from scratch.
93	Circuit bent/hacked toys and electronics, DIY circuits, prototyped and built synthesizers (kits and self-designed)
94	Modding guitars, switching pickups
96	- I play an augmented electric guitar with a midi controller embedded on it - I make my own guitar pedals that run custom software written by me - I train my own neural network models to control musical instruments or play sounds by themselves - i write max patches/other programs for live computer music performances
97	I have done some circuitbending mainly with old yamaha fm synthesizers and a yamaha drum machine.
98	Deconstructed toys to produce altered sound and function.
110	Built an AI agent to perform together live, built custom plugins and tools in C++/Faust/MaxMSP
111	I have programmed my own synths and functions in SuperCollider and Max, and designed interfaces to control parameters in TouchOSC.
112	Electric bottle for piano
117	I have built several tactile interfaces for realtime interaction with audio processes including a flex-sensitive glove, a pressure-sensitive floor pad, and a cube featuring an IMU. I have written several software programs that use these interfaces in networked performance settings.
118	Custom VSTs, Modular Stack(s), Modtracker, 2004ish Homegrown DJ Software
123	Many Eurorack modules.. Software well known in the computer music community.
129	I've coded in Max and generated acoustic models of instruments. As well as doing other things in Max with generative algorithms.
130	I built a front end so I could try the latest stable audio model in my browser. I used Deepseek on hugging face to build it with AI
131	I don't program, but have collaborated for more than 20 years with programmers to create custom built software and hardware instruments.
148	I've used a piezoelectric mic to put a DI output on a nylon string guitar. I am also researching DIY microphone and preamp building
155	custom built improvisation software in supercollider. video synthesis tools in c++, various computation systems in python
156	I have worked on the development of DYCI2.
159	I have built a specific rig with my modular synthesizer going through a seperate synthesizer as an effects unit. I also use mechanical action pipe organs in unconventional and unorthodox ways with regards to tuning and technique, often distorting the use of the organ stop and it's pulling and it's detuned effects.
165	Have built instruments out of scrap metal and wood to be hit and Amplified via contact mics.
184	Built a lyre-like instrument out of found objects and different types of strings

204	These include a sensor-laden glove, custom-built contact microphones, hydrophones, HCI devices, oscillators, effects units, VHF recorders, and inductor microphones
207	Taking a sample and playing around with its EQ settings and other various settings to reproduce a sound of my liking.
208	Added sensors to saxophones and banjos. Created instruments using digital fabrication tools and parametric design. Written many subroutines, abstractions, etc. to encapsulate code that I find useful for my work.
211	Kinect-based sound generator; hacked toy keyboard and produced different sound with Pd; similar stuff for others
212	I wrote a calculator in Python that calculates the tuning offsets needed for writing just-intoned music. I also build and modify analog electronics.
216	Hacking toys that use language.
217	Found objects, construction of bell from mixing bowl, things of that nature